StudyBuild - Project 01

# Maternal Health Risk Stratification

Health & Medical Data Science Track

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| **Project starts** | Saturday, August 22, 2026 |
| **Project work period** | August 22 - September 1, 2026 |
| **Project upload days** | Monday, August 31 & Tuesday, September 1, 2026 |
| **Final deadline** | Tuesday, September 1, 2026 |
| **Feedback, corrections & video** | Wednesday, September 2 - Friday, September 4, 2026 |

## 1. Clinical Scenario

You are working with a maternal-health analytics team that wants to better understand patterns associated with maternal risk levels. The team has a small tabular dataset containing age, blood pressure, blood sugar, body temperature and heart rate measurements together with a categorical risk label. Your task is to explore the data, identify clinically relevant patterns and build a simple baseline classifier that separates Low, Mid and High Risk groups.

**Important: this is an educational risk-stratification project, not a diagnostic tool. The goal is to practice careful health-data analysis, basic machine learning, appropriate evaluation and cautious interpretation.**

## 2. Public Dataset

UCI Maternal Health Risk is a public health dataset collected from hospitals, community clinics and maternal-health care settings in rural Bangladesh through an IoT-based risk-monitoring system. The official UCI version contains 1,013 instances, 6 input features and one categorical target, RiskLevel. UCI reports no missing values in the official dataset.

Official UCI page: https://archive.ics.uci.edu/dataset/863/pregnant+health+risk

DOI: https://doi.org/10.24432/C5DP5D

License: CC BY 4.0

Introductory paper: Review and Analysis of Risk Factor of Maternal Health in Remote Area Using the Internet of Things (IoT), 2020.

### Dataset variables

Age - maternal age in years
SystolicBP - systolic blood pressure (mmHg)
DiastolicBP - diastolic blood pressure (mmHg)
BS - blood sugar (mmol/L)
BodyTemp - body temperature (F)
HeartRate - heart rate (bpm)
RiskLevel - categorical maternal risk level

## 3. What You Should Know / Read

Python: pandas, NumPy, filtering, groupby, functions and basic plotting.
EDA: distributions, group summaries, missing/duplicate checks and outlier awareness.
Statistics: mean, median, spread, proportions and careful comparison across groups.

Visualization: Matplotlib; boxplots, histograms, bar charts and scatter plots where appropriate.
Machine learning: train/test split, Logistic Regression or Decision Tree at a beginner level, confusion matrix, Recall, F1-score and class-wise evaluation.
Health-data reasoning: false negatives, class imbalance, association vs causation, and limitations of small observational datasets.

Required tools: Python, Jupyter Notebook or Google Colab, Git/GitHub, pandas, NumPy, Matplotlib, scikit-learn.

## 4. Questions from the Clinical Analytics Team

### Q1 - What does the maternal-risk population look like?

Describe the sample and the distribution of Low, Mid and High Risk groups. Summarize the main clinical variables and identify any unusual or implausible values.

### Q2 - Which measurements differ most across risk groups?

Compare age, systolic/diastolic blood pressure, blood sugar, body temperature and heart rate across RiskLevel groups using appropriate summaries and visualizations.

### Q3 - Which variables appear most strongly associated with High Risk?

Use EDA and simple statistical comparisons to identify the variables that show the clearest separation between High Risk and the other groups. Explain that observed association does not prove causation.

### Q4 - Can a simple baseline model classify maternal risk?

Build one simple model: Logistic Regression or Decision Tree. Keep preprocessing minimal and explain every step. Do not jump directly to complex ensembles.

### Q5 - Is overall accuracy enough?

Evaluate the model with a confusion matrix and class-wise Recall/F1. Pay particular attention to the High Risk class and explain why missing a high-risk case may be more important than some other errors.

### Q6 - Where does the model fail?

Inspect misclassified examples. Identify whether errors tend to occur between Low/Mid, Mid/High or other pairs, and discuss possible overlap in the measured features.

### Q7 - Which features drive the model?

If the chosen model supports interpretable coefficients or feature importance, summarize the strongest contributors. Do not describe model importance as clinical causation.

### Q8 - What should a clinician or decision-maker take from this analysis?

Write a short, cautious interpretation of the main patterns, model performance, limitations and what additional data would be needed before any real clinical use.

## 5. Required Visualizations

Create at least five clear figures. Recommended examples: RiskLevel distribution; age by risk group; blood sugar by risk group; systolic/diastolic blood-pressure comparison; scatter plot for two clinically relevant variables; confusion matrix; and feature-importance/coefficient plot. You do not need all of these. Choose figures that actually support your analysis.

**Every figure must include: a meaningful title, readable axis labels and a written interpretation explaining what the figure shows and why it matters.**

## 6. Recommended Workflow

**Dataset understanding → quality checks → focused EDA → compare risk groups → choose a simple baseline model → train/test split → fit model → evaluate class-wise performance → inspect errors → interpret feature contributions → discuss limitations.**

## 7. Scientific & Quality Requirements

Do not treat this as a diagnostic system.
Do not report Accuracy alone. Include confusion matrix and class-wise Recall/F1, especially for High Risk.
Prevent data leakage. Fit preprocessing/modeling steps using training data only where applicable.
Use a simple model. Logistic Regression or a shallow Decision Tree is enough for Project 01.
Interpret cautiously. A feature associated with risk in this dataset is not automatically a causal risk factor.
Document limitations. Discuss sample size, geography, feature availability, label quality and lack of external validation.
Reproducibility. The notebook/code should run from beginning to end and include a fixed random seed when appropriate.

## 8. Required Deliverables

1) README.md
2) reproducible analysis/modeling notebook (.ipynb) and/or Python script (.py)
3) figures/ folder with final charts
4) short clinical-analysis summary or report
5) requirements.txt
6) original dataset source link and citation

## 9. Recommended Repository Structure

```
maternal-health-risk/
  README.md
  requirements.txt
  notebooks/analysis.ipynb
  src/model.py (optional)
  figures/
  report/summary.pdf (optional)
  data/README.md
```

## 10. README Requirements

Project title; clinical problem; dataset source and license; variable description; data-quality checks; EDA findings; model choice and justification; evaluation metrics; error analysis; feature interpretation; clinical/business implications; limitations; repository structure; and instructions for running the code.

## 11. Submission Rules - Mandatory

**Upload your project on Monday, August 31 or Tuesday, September 1, 2026.**
**Final deadline: Tuesday, September 1, 2026.**

**September 2-4 is reserved for feedback, corrections and short project-video submission.**

**ZIP/RAR files are not accepted. Submit one clean, organized GitHub folder/repository.**

**README.md and a code file are mandatory. A project containing only screenshots, charts or a final report is incomplete.**

## 12. What Success Looks Like

A strong submission should show that you can handle a small clinical dataset carefully, ask sensible questions, visualize patterns, build a simple baseline model, evaluate clinically relevant errors, interpret the model without overstating the evidence and clearly explain the limitations of the analysis.

Suggested GitHub title: Maternal Health Risk Stratification - Interpretable Machine Learning with Python